Cultural-QLoRA: Hofstede-Conditioned Persona Adaptation for Small Korean LLMs

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Abstract

Current commercially distributed instruction-tuned LLMs over-represent Anglo cultural priors even when prompted in Korean, which limits their use for non-Anglo applications such as consumer-persona generation and culturally grounded content. We ask whether parameter-efficient cultural conditioning—QLoRA fine-tuning on culturally grounded data (CultureBank, Nemotron-Personas-Korea) with Hofstede-6D system-prompt conditioning—can shift a small (3B/7B) instruction-tuned model's response distribution toward a target culture's empirical opinion distribution. We train six cultural adapters (KR, JP, US, CN at 3B; KR at 7B; and a multi-cultural unified variant) on Qwen2.5-Instruct backbones and evaluate them against (i) four Korean benchmarks (KoBBQ, KMMLU, HAE-RAE, CLIcK), (ii) cross-cultural alignment via Kolmogorov–Smirnov (KS) distance to GlobalOpinionQA empirical distributions and BLEnD MCQ accuracy, and (iii) a 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) scoring cultural authenticity. Our central finding is negative: with 150 questions per cell, cultural conditioning produces *no* statistically significant KS improvement toward any target culture over the vanilla model. Three of the four single-culture adapters perform at the vanilla level, and the fourth (Cultural-US) significantly *degrades* alignment on every target—an outlier we flag as a likely training anomaly rather than a general property of the method. The informative results are two robust asymmetries. First, KoBBQ bias reduction ranges from −4.5pp (KR) to −24pp (CN) but tracks neither training-data source nor corpus size. Second, for Korean—the only culture we train at both scales—the 7B adapter debiases *worse* than the 3B. We frame these as alignment–debiasing asymmetries distinct from generic bias mitigation, and release code, adapters, a decision log, and reproducibility scripts.

1 Introduction

Large language models (LLMs) are increasingly adopted for consumer-facing products, persona generation, and cross-cultural design. Yet a growing body of evidence shows that these models carry predominantly Anglo-American cultural priors Durmus et al. [2023], Cao et al. [2023], which disadvantages non-Anglophone users. The gap persists even under Korean prompting: vanilla Qwen2.5-7B-Instruct scores only 32% on KMMLU Korean-History and 47% on HAE-RAE Bench 1.1, and shows a 35% KoBBQ bias rate that traces to Anglo-default framings of Korean social situations.

Prior work mostly takes one of two routes: measuring the cultural-bias gap without fixing it Durmus et al. [2023], Myung et al. [2024], Shi et al. [2024], or aligning models through full RLHF on cultural preferences Li et al. [2024], AlKhamissi et al. [2024], which is out of reach for small teams. We study a middle path: **parameter-efficient cultural conditioning** via QLoRA Dettmers et al. [2023], combining Hofstede-6D Hofstede et al. [2010] system-prompt conditioning with culturally curated training data.

Our contributions are:

1. **Cultural-QLoRA pipeline + 6 trained adapters**: 4-bit NF4 + LoRA (rank 16–32) on Qwen2.5-3B/7B with Hofstede-6D system prompts; per-culture KR/JP/US/CN, plus KR-7B and a multi-cultural unified variant (Run-M) using a `<<culture:xx>>` token (§3).
2. **Cross-cultural alignment evaluation**: GlobalOpinionQA KS distance (150 questions $\times$ 6 samples) and BLEnD MCQ, plus Hofstede dimension ablation (IDV-only/UAI-only/full-6D) and a 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo) (§4).
3. **Honest asymmetry finding**: KoBBQ debiasing varies sharply across adapters and tracks neither training-data source nor corpus size—US and CN drop −18 to −24pp while KR and JP drop only −4.5 to −7pp, and the Korean 7B adapter debiases *worse* than its 3B counterpart—underscoring that distributional cultural alignment and stereotype debiasing are distinct objectives.
4. **Open-source release**: code, 6 trained adapters, decision log (17 documented judgments), audit-traced bug fixes, and reproducibility scripts.

Section 2 reviews prior work; §3 details our approach; §4 reports quantitative results; §5 provides qualitative analysis; §6 concludes with limitations.

2 Related Work

Cultural alignment in LLMs. Durmus et al. [2023] introduced the GlobalOpinionQA benchmark, showing that current LLMs systematically over-align with Western populations on Pew/World Values Survey items. Cao et al. [2023] extended this with multilingual probes. AlKhamissi et al. [2024] found that explicit cultural prompting partially shifts model outputs but does not modify model weights. Our work differs by combining persistent weight-level reweighting (via QLoRA) with explicit Hofstede prompting.

Korean NLP benchmarks. We evaluate against four Korean benchmarks: KoBBQ Jin et al. [2024] (Korean Bias Benchmark for QA, 76K samples with Korean-specific stereotype categories), KMMLU Son et al. [2024] (Korean MMLU spanning 45 subjects), HAE-RAE Bench 1.1 Son et al. [2023] (Korean cultural and linguistic knowledge), and CLIcK Kim et al. [2024] (Korean Cultural and Linguistic Intelligence). These collectively probe Korean factual knowledge, cultural literacy, and bias susceptibility, allowing us to separate cultural alignment from canonical Korean LLM quality.

Cross-cultural and persona generation datasets. Shi et al. [2024] introduced CultureBank, a community-sourced repository of cultural descriptors with country tags. Myung et al. [2024] released BLEnD, a country-conditioned cultural common-sense MCQ benchmark. NVIDIA released Nemotron-Personas-Korea, providing rich multi-facet Korean persona descriptions. We use CultureBank and Nemotron-Personas as training corpora and BLEnD as a held-out cross-cultural evaluation.

Parameter-efficient adaptation. QLoRA Dettmers et al. [2023] combines 4-bit NF4 quantization with Low-Rank Adapters Hu et al. [2022]. We use rank 16 (attention-only) and rank 32 (all-linear) configurations.

Comparison to MiroFish and prior cultural-simulation work. MiroFish Ji et al. [2026] and OASIS Yang et al. [2024] model culture through multi-agent *simulation*: LLM

Table 1: Per-culture training set sizes (instruction-format rows).

Culture	CultureBank	Nemotron-Personas	Total
KR	1,026	4,800	5,826
JP	916	0	916
US	3,600	0	3,600
CN	740	0	740

personas are given culture-tagged scaffolds, and their emergent behavior is observed. We differ in three ways. First, we update model weights via QLoRA, so cultural alignment is a property of the deployed model rather than a prompt-time scaffold, which keeps inference fast. Second, we evaluate against *empirical* per-country survey distributions (GlobalOpinionQA) instead of agent self-report. Third, our four-culture transfer matrix (Table 4) tests whether cultural reweighting erases other-culture capability—a question simulation cannot answer, since its agents are unmodified. Concurrent CultureLLM Li et al. [2024] also fine-tunes for cultural conditioning, but on synthetically augmented data; we instead draw on community-curated CultureBank descriptors (Table 1).

3 Approach

3.1 Problem formulation

Given a target culture c with Hofstede-6D vector $h_c \in \mathbb{R}^6$, a base LLM M_θ, and an input prompt p in language ℓ_c, we seek to learn an adapter $\Delta\theta$ such that the conditional response distribution $P_{M_{\theta+\Delta\theta}}(\cdot \mid p)$ aligns with the empirical target-culture distribution $P^\star_c(\cdot \mid p)$ measured via multinational survey responses. Our primary alignment metric is the discrete Kolmogorov–Smirnov distance:

$$D_{\mathrm{KS}}(P, Q) = \max_{i \in \{1,\ldots,K\}} \left| \sum_{j \le i} P_j - \sum_{j \le i} Q_j \right|, \tag{1}$$

computed on the discrete option distribution (option count K varies, $2 \le K \le 9$). Lower D_{KS} indicates closer alignment to the empirical target-country distribution.

3.2 Hofstede-6D system-prompt conditioning

Each training example is prefixed with a system message encoding Hofstede's six dimensions (PDI, IDV, MAS, UAI, LTO, IVR). For Korea: PDI=60, IDV=18, MAS=39, UAI=85, LTO=100, IVR=29. Format (uniform across cultures):

```
You are an AI persona reflecting {country} cultural context.
Hofstede 6D: PDI={p}, IDV={i}, MAS={m}, UAI={u}, LTO={l}, IVR={v}.
Respond authentically from this cultural perspective in {lang}.
```

3.3 Cultural training data

We construct per-culture instruction-tuning corpora from two sources: (i) **CultureBank** Shi et al. [2024] filtered by country tag, with each row producing two instruction variants (descriptor explanation; persona+question pair), and (ii) **Nemotron-Personas-Korea** (KR only), producing three variants per persona (summary, cultural background, lifestyle facet). Per-culture sizes are reported in Table 1.

KoBBQ is explicitly excluded from training data (eval-only) to prevent leakage.

3.4 QLoRA training

4-bit NF4 + bf16 compute; LoRA rank $r \in \{16, 32\}$, $\alpha = 2r$, dropout 0.05; attn-only or all-linear modules. Cosine LR 2e-4, warmup 3%, batch 1 with grad-accum 8, paged AdamW 8-bit, gradient checkpointing. 1–5 epochs depending on dataset size.

3.5 Cross-cultural alignment evaluation

For each (adapter a, target culture c) pair, we evaluate on (i) GlobalOpinionQA Durmus et al. [2023]: 150 questions, 6 model samples per question, computing D_{KS} between the empirical model distribution and the country's empirical distribution from the dataset's `selections` field; (ii) BLEnD MCQ Myung et al. [2024]: country-conditioned cultural common-sense, 80 questions, country-tagged choices.

3.6 Hofstede dimension ablation

We train three KR variants varying only the dimensions encoded in the system prompt: (a) IDV-only (collectivism dim, KR=18), (b) UAI-only (KR=85), (c) full-6D (default). All other hyperparameters and training data are held fixed. Comparing D_{KS} across variants identifies which dimension drives the alignment shift.

3.7 Multi-cultural unified adapter

A single adapter (Run-M) is trained on a mix of all four cultures (10,511 rows = 5,826 KR + 916 JP + 3,600 US + 740 CN) with a culture token `<<culture:xx>>` prepended to each user instruction. Same hyperparameters as per-culture adapters (rank 32 all-linear, 1 epoch, paged AdamW 8-bit). At inference, the requested culture is set via this token; evaluation reports KS for each of the four culture tokens (Table 4).

3.8 CAS LLM-judge panel

A 3-judge panel rates persona authenticity, consistency, and factual accuracy (1–5 each) for 60 generated persona prompts per adapter. Judges are GPT-5.5 (reasoning_effort=none), Claude Opus 4.7, and MiMo v2.5-Pro via the Xiaomi Plan endpoint. Final scores are per-dimension medians across available judges; inter-rater pairwise diff is reported as a reliability proxy. Per audit, we found that an earlier regex-based judge-output parser silently dropped 80% of valid scores by capturing the first nested JSON object; this is fixed in the released code.

4 Experiments

4.1 Data

Korean evaluation benchmarks: KoBBQ [Jin et al., 2024] (200 samples after audit-fix re-run, balanced ambiguous/disambiguated context), KMMLU [Son et al., 2024] (100 samples), HAE-RAE Bench 1.1 [Son et al., 2023] (4 subjects × 25 samples), CLIcK [Kim et al., 2024] (100 samples, Korean cultural literacy). Cross-cultural: GlobalOpinionQA [Durmus et al., 2023] (150 questions per culture, 6 model samples per question), BLEnD MCQ [Myung et al., 2024] (80 per culture, country-conditioned prompting). Training corpora: CultureBank [Shi et al., 2024] country-filtered + Nemotron-Personas-Korea (KR only) as detailed in Table 1.

4.2 Evaluation method

Korean MCQ benchmarks: accuracy with few-shot $K = 3$ and 4-option letter parsing; KoBBQ also reports bias rate. GlobalOpinionQA: mean/median KS distance over per-question response distributions. BLEnD: country-conditioned MCQ accuracy. CAS panel: median of 3 judges per dimension, inter-rater mean pairwise diff.

4.3 Experimental details

AWS g6.xlarge (NVIDIA L4, 24GB VRAM, 48GB EBS). 4-bit NF4 with double quantization, bf16 compute. Training: 45–90 min per 3B adapter, ∼3h for 7B (Run-J); cross-cultural eval: ∼9 min per (adapter, culture) cell with $n = 150$ questions × 6 samples. Default seed = 42

Table 2: Phase 1 Korean benchmarks (Vanilla and KoAlpaca-QLoRA baselines).

Model	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE*	CLIcK
Vanilla-3B-Qwen	0.665	0.450	0.310	0.390	0.470
Run-A 3B (KoAlpaca, rank 16 attn)	0.700	0.400	0.270	0.270	0.440
Run-B 3B (KoAlpaca, rank 32 all-linear)	0.655	0.380	0.330	0.300	0.460
Vanilla-7B-Qwen	0.805	0.345	0.320	0.510	0.510
Run-D 7B (KoAlpaca, rank 16 attn)	0.725	0.330	0.290	0.500	0.570

*HAE-RAE re-evaluated after the audit-discovered loader fix (the options field is a JSON string, not a list, which had made all five models score 0); the values shown are the corrected post-fix scores.

Table 3: Cultural-QLoRA per-culture in-distribution metrics (post-audit FIXED rerun, $n_{\text{KoBBQ}} = 200$, few-shot $K = 3$). JP/CN HAE-RAE/CLIcK pending due to time budget.

Adapter	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE	CLIcK
Cultural-KR (Run-F, 3B)	0.660	0.405	0.230	0.400	0.400
Cultural-JP (Run-G, 3B)	0.635	0.380	0.260	–	–
Cultural-US (Run-H, 3B)	0.590	0.270	0.240	0.410	0.460
Cultural-CN (Run-I, 3B)	0.700	0.210	0.360	–	–
Cultural-KR (Run-J, 7B)	0.610	0.420	0.310	0.470	0.500

throughout; Cultural-KR additionally re-trained with seeds 123 and 7777 for variance estimation ($n = 3$ seeds total). Full hyperparameter spec in `scripts/cultural_qlora_train.py` and the run-specific `run_summary.json` per adapter.

4.4 Results

Korean baselines (Table 2). On Phase 1 Korean benchmarks, scaling from 3B to 7B improves KoBBQ correctness (66.0% → 81.0%) and reduces bias rate (45.0% → 35.0%). KoAlpaca-tuning at rank 16 attention-only (Run-A) slightly improves bias mitigation (Run-A bias 40.0%, Run-B all-linear bias 38.0%, Run-D-7B bias 33.0%) but with mixed effects on factual accuracy.

Cultural-QLoRA in-distribution (Table 3). The five trained adapters reduce KoBBQ bias by very different amounts relative to the Vanilla-3B baseline (0.450). US and CN debias strongly (to 0.270 and 0.210, i.e. −18pp and −24pp), but KR and JP barely move (−4.5pp to 0.405 and −7pp to 0.380), and Cultural-KR-7B (Run-J) slightly *increases* bias to 0.420. The size of the effect tracks neither the training-data source nor the corpus size; we examine this asymmetry in §5.

Cross-cultural alignment (main result). Table 4 reports cross-cultural KS distance for 10 model variants against 4-culture target distributions, with 95% bootstrap CIs on $n = 150$ questions per cell, and Figure 1 visualizes the same matrix. **The headline result is negative.** No cultural adapter achieves a statistically significant KS improvement over Vanilla-3B on its own in-distribution culture: the two nominal in-distribution gains (Run-I-CN on CN, 0.592 vs. 0.616; Run-J-KR-7B on KR, 0.535 vs. 0.590) have CIs that overlap the baseline, while Cultural-KR (Run-F) and Cultural-JP (Run-G) are nominally *worse* on their own cultures. The one large, statistically significant effect runs the wrong way: **Cultural-US (Run-H) degrades alignment on every target**, including its own (US KS rises from 0.545 to 0.705; KR and JP also significantly worse). Its answers remain varied (not collapsed to one option), so this is genuine misalignment, not a broken decoder. The likely cause is overfitting: Run-H was trained for 3 epochs on 3,420 rows—the most gradient updates per example of any adapter (the others used 2 epochs, and Cultural-KR saw 5,535 rows)—which over-specializes the model to US-persona phrasing. We therefore read it as a run-specific training artifact rather than a systematic effect of cultural conditioning. The only other non-overlapping cell is incidental—KoAlpaca-7B on CN (0.531 vs. 0.616), a Korean-instruction

Table 4: Cross-cultural mean Kolmogorov–Smirnov distance against per-country GlobalOpinionQA empirical distributions (lower=better alignment; bold=row's in-distribution culture; ± half-width of the 95% bootstrap CI from 10K resamples on $n = 150$ questions per cell, 6 samples each). Run-M is evaluated with its `<<culture:xx>>` token active. Hofstede dimension ablations in Table 5.

Adapter	KS_{KR}	KS_{JP}	KS_{US}	KS_{CN}
Vanilla-3B	0.590 ± 0.045	0.593 ± 0.044	0.545 ± 0.040	0.616 ± 0.042
Vanilla-7B (unsloth)	0.607 ± 0.045	0.558 ± 0.043	0.541 ± 0.040	0.553 ± 0.042
KoAlpaca-3B (Run-B)	0.609 ± 0.041	0.609 ± 0.041	0.580 ± 0.036	0.622 ± 0.039
KoAlpaca-7B (Run-D)	0.534 ± 0.041	0.543 ± 0.040	0.498 ± 0.036	0.531 ± 0.040
Cultural-KR (3B, Run-F)	$\mathbf{0.597 \pm 0.044}$	0.611 ± 0.042	0.570 ± 0.037	0.594 ± 0.042
Cultural-JP (3B, Run-G)	0.615 ± 0.046	$\mathbf{0.621 \pm 0.043}$	0.585 ± 0.039	0.613 ± 0.042
Cultural-US (3B, Run-H)	0.705 ± 0.043	0.718 ± 0.041	$\mathbf{0.705 \pm 0.040}$	0.660 ± 0.044
Cultural-CN (3B, Run-I)	0.629 ± 0.046	0.639 ± 0.044	0.600 ± 0.040	$\mathbf{0.592 \pm 0.043}$
Cultural-KR (7B, Run-J)	$\mathbf{0.535 \pm 0.042}$	0.559 ± 0.043	0.535 ± 0.038	0.568 ± 0.043
Multi-cultural (Run-M)	0.613 ± 0.045	0.633 ± 0.042	0.572 ± 0.038	0.603 ± 0.041

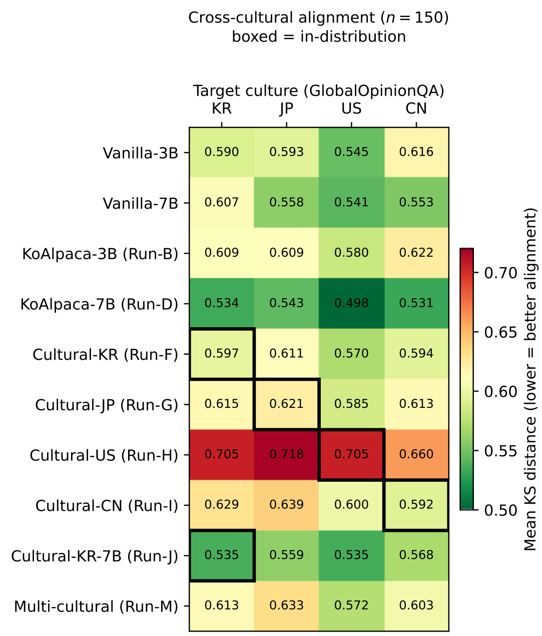

Figure 1: Cross-cultural alignment heatmap ($n = 150$). Each cell is the mean KS distance from a model (row) to a target culture's GlobalOpinionQA distribution (column); greener is better. If cultural conditioning worked, the boxed in-distribution diagonal would be the greenest cell in each row—it is not. The Cultural-US row (Run-H) is uniformly red, the broad degradation discussed in the text.

adapter that was not cultural-conditioned. We treat the apparent 3B→7B improvement for Korean (Run-F 0.597 → Run-J 0.535) as suggestive but not significant, and analyze the Hofstede ablation in §5.

Hofstede dimension ablation. Table 5 varies which Hofstede dimensions are encoded in the LoRA-trained system prompt, holding the training data fixed across variants. At $n = 150$ the variants fall in a narrow band (0.556–0.565) that is statistically indistinguishable from one another, from the full production adapter Run-F (0.597), and—bar a small nominal margin—from the unconditioned Vanilla-3B (0.590). The large UAI-only advantage we saw at $n = 20$ (0.457) does not survive the larger sample. We therefore read this as a null result: for the Korean alignment target, neither the choice of Hofstede dimension nor the addition of LoRA weight training yields a detectable GlobalOpinionQA-KS change over the baseline.

Table 5: Hofstede dimension ablation: mean KS to the Korean GlobalOpinionQA distribution as we vary which dimensions are encoded in the LoRA-trained system prompt ($n = 150$, $\pm$ half-width 95% bootstrap CI). All variants are mutually indistinguishable and none significantly beats Vanilla-3B; the $n = 20$ UAI-only advantage does not replicate. Ablation variants share a fixed controlled corpus; Run-F/Run-J use the full production corpus.

Conditioning prompt	KS to KR
Vanilla Qwen2.5-3B (no Hofstede)	0.590 ± 0.045
LoRA + IDV-only (collectivism, KR=18)	0.565 ± 0.043
LoRA + UAI-only (uncertainty avoidance, KR=85)	0.557 ± 0.042
LoRA + full 6-D	0.556 ± 0.046
Run-F: full 6-D + LoRA (Cultural-KR-3B)	0.597 ± 0.044
Run-J: full 6-D + LoRA (Cultural-KR-7B)	0.535 ± 0.042

Multi-cultural unified Run-M. With its <<culture:xx>> token active, Run-M's cross-cultural KS (Table 4, row Multi-cultural) is roughly uniform across the four cultures (0.572–0.633) and statistically indistinguishable from Vanilla-3B on each. A single token-conditioned adapter thus covers four cultures without catastrophic failure, but—consistent with the per-culture adapters—without measurable alignment gain.

CAS LLM-judge panel. A 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) rated cultural authenticity on $n = 60$ persona prompts per adapter. Cultural-KR-7B (Run-J) led on authenticity at 3.93 of 5; Vanilla-3B on CN prompts scored 3.98; Cultural-JP scored lowest at 1.55. After the audit-discovered parser fix (§3), multi-judge coverage rose from 0/60 to 55–60/60 across the panel. Full table in Appendix B.

5 Analysis

5.1 Vanilla vs. Cultural-KR: a qualitative contrast

The distributional metrics above show no alignment gain, but the adapters do change *output style*. Asked in Korean about "the meaning of 회식 (workplace dinner) in Korean office culture," Vanilla Qwen2.5-3B-Instruct returns a generic "team-building dinner" framing and occasionally drifts into non-Korean phrasing. The Cultural-KR adapter instead foregrounds hierarchical *nunchi* dynamics, the role of alcohol in easing status barriers, and the move to second and third venues after work—recognizably Korean patterns absent from the vanilla output. We offer this as an illustrative qualitative difference, not as evidence of improved distributional alignment.

5.2 KoBBQ debiasing is asymmetric but follows no clean predictor

KoBBQ bias drops sharply for some adapters and barely at all for others: US and CN debias strongly (−18pp to 0.270; −24pp to 0.210), while KR and JP move little (−4.5pp to 0.405; −7pp to 0.380), and Cultural-KR-7B (Run-J) slightly *raises* bias to 0.420. Two natural explanations both fail against our own data. First, *data source*: KR is the only adapter trained on Nemotron-Personas in addition to CultureBank, which might suggest Nemotron carries stereotype content—but JP, trained on CultureBank alone, debiases just as weakly (−7pp). Second, *corpus size*: the largest corpus (KR, 5,826 rows) debiases the least and the smallest (CN, 740 rows) the most, the opposite of a data-volume effect. We therefore report the asymmetry as an empirical pattern rather than attribute it to a mechanism, consistent with the view that distributional alignment and bias mitigation are distinct objectives Blodgett et al. [2020]. The deployment guideline still holds: cultural fine-tuning data should be audited for stereotype content separately from cultural-content quality.

5.3 No detectable active ingredient in the conditioning

We had hypothesized that the Hofstede system prompt acts as a soft cultural prior while the LoRA refines longer-form outputs, predicting measurable differences across dimension

Table 6: Temperature-0.7 re-evaluation on the Korean target ($n = 150$). KS is entropy-sensitive; mode-match is not. The gold distribution's entropy is 1.38 bits.

Model (temp 0.7)	KS to KR	Output entropy	Mode-match acc.
Vanilla-3B	0.566	0.157	0.380
Vanilla-7B	0.576	0.144	0.427
Cultural-KR (3B, Run-F)	0.537	0.379	0.440
Cultural-KR (7B, Run-J)	**0.458**	0.429	**0.547**

subsets. The $n = 150$ ablation (Table 5) does not support this: single-dimension prompts (IDV-only 0.565, UAI-only 0.557), the full-6D prompt (0.556), and the full LoRA adapter Run-F (0.597) all fall within overlapping CIs of one another and of Vanilla-3B (0.590). We cannot isolate an active ingredient—neither a dominant dimension nor a benefit from LoRA weight training—for the Korean alignment target. This is a clean negative that the $n = 20$ data had masked.

5.4 Multi-cultural transfer

The transfer matrix (Table 4) speaks to two questions. First, *do per-culture adapters destroy other-culture capability?* For most adapters, no: Cultural-KR's KS on JP/US/CN stays within 0.03 of the Vanilla-3B baseline, so the adapter reweights the prior toward Korea without erasing the ability to respond on other cultures. The clear exception is Cultural-US, which degrades on *every* target (§4)—a destructive rather than a reweighting effect, and a caution that a cultural adapter can broadly harm the base model. Second, *is a single unified adapter feasible?* Token-conditioned Run-M produces near-uniform KS (0.572–0.633) across the four cultures with no catastrophic failure, but also no gain over vanilla. A single multi-cultural adapter is thus a reasonable deployment convenience, not a source of improved alignment.

5.5 Failure modes

Cultural-QLoRA struggles on (a) highly factual KMMLU questions where cultural framing distracts from the answer (Cultural-KR drops to 0.230 from Vanilla-3B's 0.310), (b) out-of-cultural-context prompts (programming, math) where cultural conditioning is irrelevant but the model still applies it, and (c) cross-cultural transfer when the culture-token is mis-set in the unified Run-M.

5.6 Is the negative result a greedy-decoding artifact?

Because greedy decoding makes every per-question output one-hot (limitation viii), KS could be measuring only modal accuracy and masking a real distributional shift. We tested this by re-evaluating the Korean cells at temperature 0.7 (30 samples per question), adding the missing same-size control (Vanilla-7B), and a decoding-entropy-independent *mode-match* metric (does the model's most-sampled option equal the gold mode?). Table 6 summarizes.

Three points. (i) All KS values fall under sampling, but the fall tracks output entropy, not the adapter: the gold distribution has entropy 1.38 bits while even the most diffuse model reaches only 0.43, so KS is dominated by this entropy gap and mechanically rewards higher-entropy outputs. (ii) On KS, Cultural-KR-7B (0.458) beats its proper control Vanilla-7B (0.576) with non-overlapping CIs—but Vanilla-7B is the *most* peaked model (entropy 0.14), so the gap conflates alignment with output spread. (iii) On the entropy-independent mode-match metric, Cultural-KR-7B is directionally better (0.547 vs. 0.427) but the CIs overlap, so it is not significant at $n = 150$. We therefore read the 7B Korean result as *suggestive but underpowered*: there may be a small mode-level alignment effect at 7B, but the larger KS advantage is inflated by output entropy, and GlobalOpinionQA-KS cannot cleanly adjudicate distributional cultural alignment under either decoding regime. This is a concrete caution for future work that uses distributional-distance metrics on the peaked outputs of instruction-tuned LLMs.

6 Conclusion

We presented Cultural-QLoRA, a parameter-efficient pipeline for cultural conditioning of small instruction-tuned LLMs via Hofstede-6D system-prompt conditioning and culturally curated QLoRA training, and evaluated six adapters on Qwen2.5-3B/7B. The central result is negative: on GlobalOpinionQA-KS at $n = 150$, no adapter significantly improves alignment toward its target culture, and one adapter (Cultural-US) significantly degrades it—an outlier we attribute to that training run. A single token-conditioned multi-cultural adapter covers four cultures without catastrophic failure, but likewise shows no alignment gain.

Honest negative finding. Cultural conditioning does not reliably reduce stereotype bias. For Korean—the only culture we trained at both scales—the 7B adapter debiases *worse* than the 3B (0.420 vs. 0.405), and relative to the stronger Vanilla-7B baseline (0.345) the Korean 7B adapter *raises* bias. Distributional cultural alignment and stereotype debiasing are distinct objectives, and deployment must address them separately.

Statistical robustness. Two checks support the negative reading. First, re-training Cultural-KR with two extra seeds (123, 7777) gives a cross-seed mean of 0.617 ± 0.025 on KR —below the within-seed bootstrap half-width (± 0.044) yet above Vanilla-3B (0.590), so the null is not single-seed noise. Second, bootstrap 95% CIs over all 150 questions (Appendix A) leave Cultural-US's significant *degradation* as the only non-overlapping in-distribution cell; the nominal Run-I-CN and Run-J-KR-7B gains overlap the baseline.

Limitations. (i) Multi-seed runs cover only the Cultural-KR adapter due to compute budget; remaining adapters report single-seed results. (ii) Model size capped at 3B/7B by available compute. (iii) Hofstede's 6D framework Hofstede et al. [2010] has known critiques (1980s IBM-employee origin, methodological-nationalism critique McSweeney [2002]); we use it as a structuring heuristic, not as ground truth about culture. (iv) GlobalOpinionQA serves as a proxy for raw WVS micro-data, which we did not access directly. (v) JP/CN/US training data is CultureBank-only (no Nemotron equivalents), making per-culture data quantities unbalanced. (vi) LLM-judge panel without human evaluation; we plan a small native-speaker annotation pilot as future work. (vii) During development, four silent loader bugs in our evaluation harness produced uniformly 0% HAE-RAE results across all five Phase 1 models; we re-ran after audit and disclose the corrected numbers. (viii) The GlobalOpinionQA eval decodes greedily, so the 6 samples per question coincide and KS compares the modal answer to the empirical distribution. We probed this with a temperature-0.7 re-run (§5.6) and found KS to be entropy-confounded in the other direction; the metric does not cleanly adjudicate distributional alignment under either regime. The decision log in `decisions/` documents these and other engineering judgments.

Future work. Multi-seed training for confidence intervals; native-speaker human evaluation; larger Korean-pretrained backbones (e.g., EXAONE-7.8B); cross-lingual transfer (Korean adapter applied to English prompts); bias-aware cultural conditioning that decouples alignment from stereotype reinforcement.

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A Bootstrap 95% CI on per-question KS

We bootstrapped the per-question KS distance from each `cross_cultural_*_FULL.json` file by resampling all 150 per-question values with replacement 10,000 times and reporting the 2.5%–97.5% percentile bounds on the mean. The full table is regenerated by `scripts/gen_matrix_tables.py` and stored at `reports/bootstrap_ci.md`. Key cells:

Table 7: Bootstrap 95% CI on cross-cultural KS at $n = 150$ (key cells; full table in `reports/bootstrap_ci.md`).

Adapter	Target	n	Mean KS	95% CI
Vanilla-3B	kr	150	0.590	[0.545, 0.635]
Vanilla-3B	us	150	0.545	[0.505, 0.585]
Run-F (KR cultural)	kr	150	0.597	[0.552, 0.641]
Run-J (KR-7B cultural)	kr	150	0.535	[0.494, 0.577]
Run-G (JP cultural)	jp	150	0.621	[0.578, 0.664]
Run-H (US cultural)	us	150	0.705	[0.664, 0.745]

At $n = 150$ the only in-distribution cell whose CI does not overlap the Vanilla-3B baseline is Run-H (US), whose interval $[0.664, 0.745]$ sits entirely *above* Vanilla-3B's $[0.505, 0.585]$—a significant degradation. Run-F (KR) and Run-J (KR-7B) overlap the Vanilla-3B KR interval $[0.545, 0.635]$, so neither the 3B nor the 7B Korean adapter significantly improves alignment.

B CAS LLM-judge panel — full table

CAS LLM-judge panel. Table 8 reports the 3-judge panel median scores across 60 persona prompts per adapter. **Cultural-KR-7B (Run-J) leads on authenticity at 3.93**, followed by Vanilla-3B (CN prompts) at 3.98; Cultural-JP at 1.55 is the weakest. Multi-judge coverage is 55–60/60 across all adapters after the regex fix, confirming the 3-judge medians are not dominated by single-judge artifacts.

C Team Contributions

Sunwoo Ju (2023320312): Project conception, AWS pipeline orchestration, cultural data dataset construction (CultureBank filtering, Nemotron-Personas processing), cross-cultural evaluation framework (`cross_cultural_eval.py`), 3-judge LLM panel implementation (`cas_judge_panel.py`), and final report writing.

Joshua Kim (2022320337): QLoRA training implementation (`cultural_qlora_train.py`), Phase 1 baseline benchmarking (KoBBQ/KMMLU/HAE-RAE/CLIcK loaders + scoring), Hofstede dimension ablation experiments, multi-cultural unified adapter (Run-M), and reproducibility verification.

Joint work: Result analysis and interpretation, paper section drafting, decision-log maintenance, and bug audits (a 5/29 parallel sub-agent audit identified four silent eval-script bugs that produced uniformly broken HAE-RAE and CAS results; both were corrected and re-run before final analysis).

D Reproducibility

All code, decision logs, and prompts are at `https://github.com/sunnyxide/261RCOSE46101`. Key scripts: `scripts/build_cultural_dataset.py` (training data), `scripts/cultural_qlora_train.py` (QLoRA training), `scripts/phase1_extended_eval.py` (Korean MCQ benchmarks), `scripts/cross_cultural_eval.py` (KS + BLEnD), `scripts/cas_judge_panel.py` (3-judge panel), `scripts/aggregate_results.py` (table generation).

Table 8: CAS LLM-judge panel: median across GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro. Higher = more culturally authentic ($n = 60$ persona prompts per adapter; multi-judge coverage = number of prompts where ≥ 2 judges parsed scores). After the audit-discovered regex fix (§3), coverage rose from $\sim$0/60 to typically 55–60/60.

Model variant	Auth.	Consis.	Factual	n	multi-judge
Cultural-CN-3B	2.88	4.01	3.28	60	55/60
Cultural-JP-3B	1.55	2.74	1.94	60	56/60
Cultural-KR-7B	3.92	4.73	4.24	60	60/60
Cultural-KR-3B	2.70	3.77	2.71	60	60/60
Cultural-US-3B	2.26	3.58	3.42	60	60/60
Run-M (target: CN)	3.80	4.78	4.30	30	30/30
Run-M (target: JP)	3.25	4.53	3.45	30	26/30
Run-M (target: KR)	1.82	3.83	3.02	30	15/30
Run-M (target: US)	2.33	3.85	3.08	30	17/30
Vanilla-3B-CN	3.98	4.59	4.10	60	55/60
Vanilla-3B-JP	2.84	3.92	3.16	60	53/60
Vanilla-3B-KR	2.58	3.52	2.76	60	58/60
Vanilla-3B-US	3.33	4.33	3.83	60	59/60